PHASE 8 OF 8

Project Demonstration

Presenting the complete LegalEase workflow

Project: LegalEase – AI-Powered Legal Document Generator

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8.1 Purpose of this Phase

The Project Demonstration phase presents the finished LegalEase application. The team shows the complete workflow, from starting the servers to downloading the final document, and explains the architecture and technologies that make it work. The demonstration proves that the requirements were met and that the components work together.

8.2 Pre-Demonstration Checklist

- Python environment and all dependencies are installed.
- The Gemini API key is present in the .env file and the internet connection is working.
- Sample inputs (for example a Lease Agreement or Employment Contract) are ready.
- A text editor, Word-compatible viewer and PDF reader are available to open the downloads.
- Each team member knows which part of the demonstration they will present.

8.3 Demonstration Steps

Step	Action	What the audience sees
1	Start the FastAPI backend	Server running and ready for requests
2	Start the Streamlit frontend	Application launches in the browser
3	Open the LegalEase interface	Input form is displayed
4	Select a document type such as Lease Agreement or Employment Contract	Document type is set
5	Enter parties, terms, effective date and jurisdiction	Form is completely filled
6	Click Generate Document	Request is sent; progress is shown
7	Show the generated preview	Structured draft appears
8	Edit the content if required	Changes appear in the preview
9	Download as TXT, DOCX and PDF	Three files are saved

Step	Action	What the audience sees
10	Explain the architecture and technologies used	Streamlit, FastAPI, Gemini and formatters explained

8.4 Sample Demonstration Scenario

A sample scenario is a Lease Agreement between two parties with a stated rent, duration and obligations, an effective date and a jurisdiction. After generation, the presenter edits one clause in the preview to show that changes are kept, and then downloads the DOCX and PDF versions and opens them to show the headings, terms, logo and footer.

8.5 Talking Points for the Architecture Explanation

- **Frontend (Streamlit):** collects inputs, previews, edits and provides downloads.
- **Backend (FastAPI):** the /generate endpoint validates requests with Pydantic and returns responses.
- **AI core (Gemini):** creates the document text from the inputs.
- **Formatting modules:** python-docx and FPDF create the Word and PDF files; TXT gives plain text.
- **Data flow:** User Input to Streamlit to FastAPI /generate to Gemini to Generated Text to Preview/Edit to export.

8.6 Anticipated Questions

Question	Suggested answer
Is the generated document legally valid?	It is a draft to be reviewed appropriately before actual legal use.
Why separate the frontend and backend?	Keeps the AI service reachable through an API and each part easy to test and change.
Can the text be changed?	Yes. The preview is editable and exports use the edited text.
Which formats are supported?	TXT, DOCX and PDF.

8.7 Limitations and Future Scope

- Draft quality depends on the AI model and the information supplied.
- Requires an internet connection and a valid Gemini API key.

- Possible future work: more document types, saved drafts, user accounts, and improved templates and branding options.

8.8 Conclusion

LegalEase demonstrates how Generative AI can be integrated with a web application to simplify legal document drafting. It combines Streamlit, FastAPI, Gemini and document-generation libraries to provide a workflow for generating, reviewing, editing and exporting structured legal document drafts.

Project team: Ganesh Prabhu, Gopinath, Hariharan, Gowtham.

8.9 Demonstration Roles

Part	Content
Introduction	Problem statement and objective of LegalEase
Live workflow	Start servers, fill form, generate, preview, edit
Export	Download TXT, DOCX and PDF and open them
Technical explanation	Architecture, data flow and technologies
Closing	Limitations, disclaimer, future scope and questions

8.10 Success Criteria for the Demonstration

- The complete workflow runs from input to download without manual workarounds.
- All three export formats open correctly.
- The team can clearly explain each component and the data flow.
- The disclaimer about reviewing generated drafts is stated.

8.11 Contingency Plan

If the network or the Gemini service is unavailable during the demonstration, the team can show previously generated and downloaded documents and explain the workflow using the architecture diagram, then repeat the live generation when the connection is restored.

8.12 Key Messages to Leave with the Audience

- Generative AI can reduce the effort of producing a first draft of a legal document.
- A clean separation of Streamlit, FastAPI, Gemini and formatters makes the system simple to understand and extend.
- The user stays in control: every draft can be reviewed and edited before export.

- Generated documents are drafts and must be reviewed appropriately before actual legal use.

8.13 Follow-up Actions After the Demonstration

1. Note questions and feedback from evaluators.
2. Record any improvements suggested for future versions.
3. Finalize and submit the project documents.
4. Archive the project files, including the setup instructions.